

Event-Driven Telemetry Platform – Project Timeline

This document outlines a realistic, resume-oriented timeline for building the Event-Driven Telemetry Platform. The focus is on backend system design, correctness, and clarity rather than speed.

Phase	Duration	Goals	Key Deliverables
Phase 0: Infrastructure	1-2 days	Set up local infrastructure and validate Docker Compose, Kafka (KRaft), topic creation, C	
Phase 1: Ingestion Service	2-4 days	Build stateless HTTP service that publishes telemetry events	FastAPI endpoints, schema validation, auth n
Phase 2: MongoDB Consumer	2-3 days	Persist raw telemetry events reliably	Kafka consumer, idempotent MongoDB writes, ind
Phase 3: Redis Aggregator	2-3 days	Maintain real-time metrics and counter	Redis consumer, TTL-based metrics, active users
Phase 4: Metrics API	1-2 days	Expose telemetry data via HTTP APIs	GET /metrics endpoints reading from Redis
Phase 5: Simulator & Load	1-2 days	Demonstrate system behavior under load	Event simulator script, basic stress testing
Phase 6: Polish & Docs	1-2 days	Make project resume-ready	README, architecture docs, demo instructions

Total Estimated Time: ~12–16 focused days. The project can comfortably be completed in 3–4 weeks alongside interviews or a job.

This timeline is intentionally structured to show architectural maturity, not just feature completion. Each phase builds a complete vertical slice of the system.